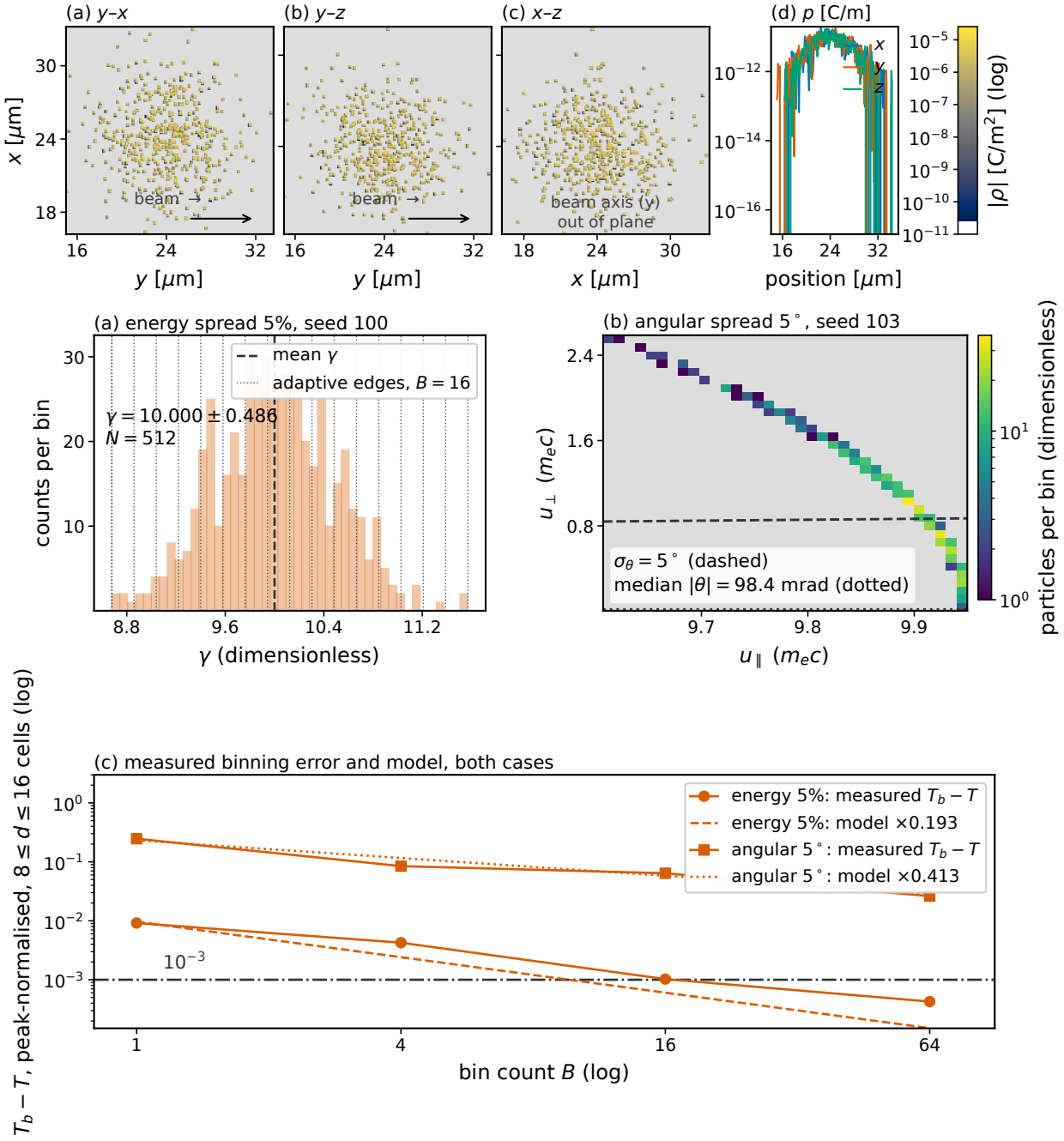

(a) y–x
(b) y–z
(c) x–z
(d) p [C/m]
x [μm]
y [μm]
beam →
beam axis (y)
out of plane
position [μm]
|ρ| [C/m²] (log)
(a) energy spread 5%, seed 100
mean γ
adaptive edges, B = 16
γ̄ = 10.000 ± 0.486
N = 512
counts per bin
γ (dimensionless)
(b) angular spread 5°, seed 103
u⊥ (mₑc)
u∥ (mₑc)
σθ = 5° (dashed)
median |θ| = 98.4 mrad (dotted)
particles per bin (dimensionless)
(c) measured binning error and model, both cases
energy 5%: measured Tb − T
energy 5%: model ×0.193
angular 5°: measured Tb − T
angular 5°: model ×0.413
10⁻³
bin count B (log)
Tb − T, peak-normalised, 8 ≤ d ≤ 16 cells (log)